Place Value & Expanded Form — 7-Digit Numbers

Number discs + place-value charts (up to millions)

ID 118255 | Subject Maths | Grade Five | Time 40 min | Type Pictorial & Abstract

Objectives

2 min

- Students will write 7-digit numbers in expanded form using number discs and place-value charts. Analyse

Resources & Support

Resources

- Whiteboard and markers
- Student notebooks
- A 7-column place-value chart (Millions ... Ones)

Misconception alert — missing zeros

Students often miss zeros in expanded form (e.g. writing 20,000 instead of 200,000) because they lose track of the place-value columns. Correction: "Look at our drawing — it shows 2 discs in the Hundred Thousands column, and each disc is worth 100,000, making it 200,000."

Introduction

5 min

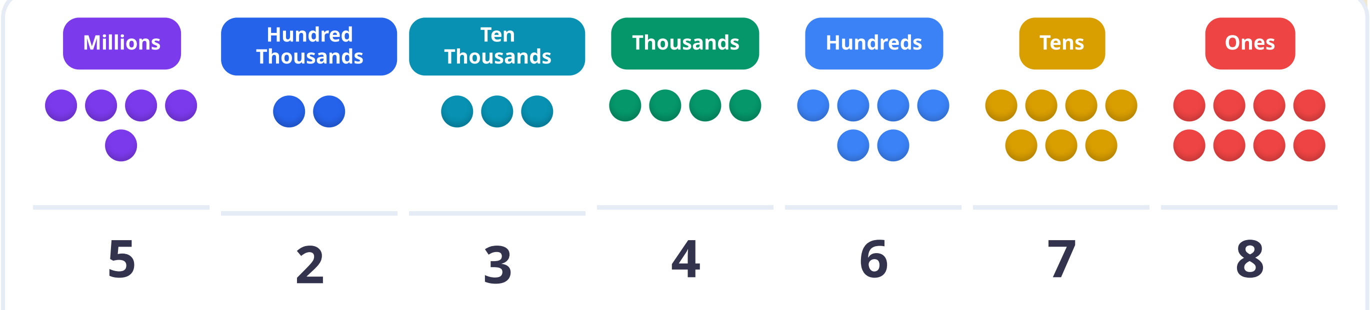
Today we learn to break 7-digit numbers into expanded form using number discs — so we can read large amounts, like the cost of building a new hospital in our city.

Recall: Remember the base-ten blocks from yesterday? How can we show that as a simple drawing today? — We can draw circles or dots inside a place-value chart.

Try it: Imagine Ali has 3,000,000 Rupees. In the air, draw the discs for this number in the Millions column. Ready? Go!

The board — 5,234,678 as discs + expanded form

3 min



5,000,000 + 200,000 + 30,000 + 4,000 + 600 + 70 + 8 = 5,234,678

Each ● is one unit of its column (e.g. one hundred thousand = 100,000).

Explanation & Teaching

15 min

Look (45s) — Give students 45 seconds to study the chart and think about what it represents.

Count each column — With Bunty, count the discs column by column: Millions → 5,000,000; Hundred Thousands → + 200,000; Ten Thousands → + 30,000; Thousands → + 4,000; and so on — each disc translates to the number with exactly the right zeros.

Combine — Add all the expanded parts together to get the standard form: 5,234,678.

Check for understanding

- Which part of the drawing shows 200,000? — The 2 discs in the Hundred Thousands column.
- If Bunty adds one more disc to the Thousands column, what does that column become? — 5,000 instead of 4,000.
- How does drawing discs help avoid missing zeros? — You read the exact place-value heading, so you know how many zeros belong to that column.

Guided Practice

17 min

In pairs, one student is the "drawer" and the other the "writer". The drawer draws the number discs for 6,781,234 in a place-value chart; the writer writes the expanded-form equation underneath. Then switch roles for 3,210,456.

Teacher note

Individual (Page 6): imagine counting bricks to build a new school in Lahore — draw the discs and write 7,654,321 in expanded form.

For struggling students

Provide a pre-drawn 7-column place-value chart for students to draw their discs into.

For advanced students

Write a number greater than 7,654,321 and prove it by drawing the discs for the Millions and Hundred Thousands columns.

Assessment & Wrap-up

3 min

Thumbs up for True, thumbs down for False.

- 3 discs in the Hundred Thousands column equals 30,000. 👎
- Expanded form helps us see the value of each digit in a big number. 👍
- 7,000,000 has six zeros. 👍
- If I have 0 discs in the Tens column, I write nothing for the Tens place in my standard number. 👎

Exit ticket

Ask: Where else in your neighbourhood do you see numbers like this? (property prices, bank boards, population signs).